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RESEARCH ARTICLE

Domain-Adapted Retrieval-Augmented Generation for Technical Documentation: Enhancing Reliability and Faithfulness in Technical QA

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ABSTRACT This paper introduces a domain-adapted Retrieval-Augmented Generation (RAG) pipeline specifically designed for technical question answering within industrial documentation. To overcome the limitations of generic retrieval systems in processing version-sensitive and semantically dense content, we develop a structured bilingual corpus of 20,000 manuals and fine-tune a dual-mode embedder (BGE-M3) with triplet loss and hard negatives. We further incorporate a two-stage reranking mechanism and an LLM-based query auditing module (LLM-Critic) to enhance retrieval precision and contextual adequacy. Experimental evaluation demonstrates substantial improvements in Recall@5 (from 0.67 to 0.86) and MRR (from 0.52 to 0.80), confirming the effectiveness of layered retrieval optimization in reducing ambiguity, strengthening grounding, and mitigating hallucinations. We have identified further tasks to address remaining challenges in managing multimodal content, resolving version disambiguation, and ensuring scalability for real-world deployment.

INDEX TERMS LLM, RAG, question answering, technical, reliablitty, faithfulness, domain adaptation.

I. INTRODUCTION

Technical documentation, encompassing equipment manuals, engineering standards [1], [2], [3], and regulatory guidelines [3], [4], [5], poses distinctive challenges for automated language processing. In contrast to general-domain text, technical documents undergo frequent revision, employ highly specialized terminology, and adhere to rigid structural formats with extensive cross-references and version dependencies. These characteristics constrain the effectiveness of traditional rule-based systems and ontology-driven pipelines, which, despite their precision, require substantial manual effort and lack adaptability. Similarly, the limited availability of high-quality in-domain training data and the limited resilience to frequent documentation updates constrain domain-specific fine-tuning (DFT) of language models.

Retrieval-Augmented Generation (RAG) has emerged as a promising alternative that integrates document retrieval with generative modeling, thereby providing answers grounded in retrieved authoritative document fragments [6], [7], [8], [9], [10], [11]. Relative to DFT, RAG enhances factual accuracy, mitigates hallucination risk via retrieval grounding, rather than eliminating it entirely, and enables transparent source attribution [12], [13] — capabilities that are particularly useful for technical and safety-sensitive applications [10]. Nevertheless, RAG systems remain insufficiently developed for technical domains, where heterogeneous document structures, evolving terminology, and recurrent near-duplicate fragments pose persistent challenges [14], [15], [16], [17], [18]. Addressing these challenges is essential for deploying retrieval-grounded technical QA over industrial

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documentation, where consistency, traceability, and version disambiguation are critical [19], [20], [21].

This paper presents a domain-adapted RAG architecture for technical documentation and empirically shows that targeted retrieval-level adaptation — via retrieval strategy comparison, hard-negative fine-tuning, and multi-stage reranking — substantially improves retrieval fidelity, disambiguation accuracy, and contextual grounding, thereby increasing factual reliability in technical QA.

A key requirement for transparent RAG benchmarking in industrial technical QA is consistent latency reporting under a fixed hardware and batch-inference protocol. In this study, latency values are reported in milliseconds (ms) and measured under controlled batch conditions on a single inference server (hardware and batch setup will be detailed in Section III). Baseline comparisons are restricted to fully reproducible system ablations to avoid unverifiable approximations.

II. RELATED WORK

Retrieval-Augmented Generation (RAG) has emerged as a prominent approach for enhancing factual consistency in language models, particularly for knowledge-intensive tasks. Initially introduced by Lewis et al. (2020), RAG combined neural retrievers with sequence-to-sequence generators and demonstrated strong performance in open-domain QA using Wikipedia [6]. Subsequent research extended this paradigm to specialized domains, including biomedical and legal corpora to technical documentation — by incorporating domain-adaptive retrieval and fine-tuning strategies [20], [21], [22], [23], [24], [25].

Despite encouraging results, applying RAG to technical domains such as industrial manuals, engineering specifications, and enterprise support documents entails unique challenges. These challenges include semantic ambiguity caused by overlapping terminology, redundancy arising from near-duplicate document versions, and evolving structural formats that compromise retrieval performance. Initial attempts to address these limitations employed hybrid retrievers [26] and structured knowledge integration [27], [28], and demonstrated limited scalability and inconsistent robustness under frequent documentation updates [29]; however, these approaches frequently lacked unified latency reporting and sufficient in-domain baseline reproducibility, which reduces the transparency of comparative gains. Recent studies have identified retrieval optimization — not just better generation — as the critical bottleneck in domain-specific RAG systems. Empirical analyses in domains such as law, medicine, and aviation consistently attribute factual errors to irrelevant or misaligned retrieval [30], [31]. In response, several strategies have been proposed, including semantic chunking, dense-sparse retrieval ensembles, and two-stage reranking pipelines. These approaches are designed to improve top-k precision while maintaining acceptable latency, a property particularly critical in technical QA where traceability and factual grounding are essential.

Although reranking and contrastive learning have improved retrieval precision, many methods remain dependent on large annotated datasets or handcrafted ontologies, limiting portability. Additionally, contrastive or end-to-end training approaches often suffer from overfitting to narrowly defined schemas. In parallel, hallucination mitigation strategies — such as context prepending and agent-based reranking — have demonstrated potential, yet remain difficult to evaluate consistently due to fragmented benchmarks and heterogeneous evaluation metrics.

Taken together, these reviewed works underscore the central importance of retrieval precision for deploying RAG systems in technical domains. Although prior work has introduced valuable components — such as hybrid retrievers, domain adaptation, and reranking — these techniques are rarely integrated into cohesive architectures capable of addressing the complexity of industrial documentation. To address this gap, we present a domain-adapted RAG pipeline that systematically improves retrieval fidelity through comparative strategy evaluation, hard-negative fine-tuning, and multi-stage reranking. Experimental results demonstrate that this integrated approach achieves measurable improvements in factual accuracy, semantic disambiguation, and contextual alignment for technical QA.

III. METHOD

In technical QA tasks, users frequently require highly specific, version-sensitive information from complex manuals, including configuration parameters, component tolerances, or troubleshooting procedures. Such scenarios necessitate not only factual correctness but also precise contextual alignment with the relevant hardware variant or operational constraint. To address this challenge, we propose a domain-adapted RAG pipeline that performs retrieval optimization across multiple stages.

To support domain-aware retrieval, we built a corpus of industrial technical manuals in the HVAC, automation, and electrical systems domains. Documents were segmented into semantically coherent, hierarchy-preserving chunks to enable accurate context retrieval. Subsequently, dense, sparse, and hybrid retrieval strategies were benchmarked under domain-specific conditions capturing the unique characteristics of technical documentation, such as near-duplicate version fragments, hierarchical formatting, and specialized terminology. Based on these findings, the best-performing embedder is fine-tuned using triplet loss with hard negatives to enhance disambiguation of semantically similar fragments. Retrieved candidates were then subsequently passed through a two-stage reranking pipeline, and an LLM-Critic module was introduced to audit and reformulate underspecified queries. This multi-stage architecture incrementally improves retrieval precision, contextual adequacy, and factual faithfulness in technical QA.

The query auditing module (LLM-Critic) evaluates the sufficiency of retrieved context and reformulates underspecified queries when necessary, thereby mitigating hallucinations

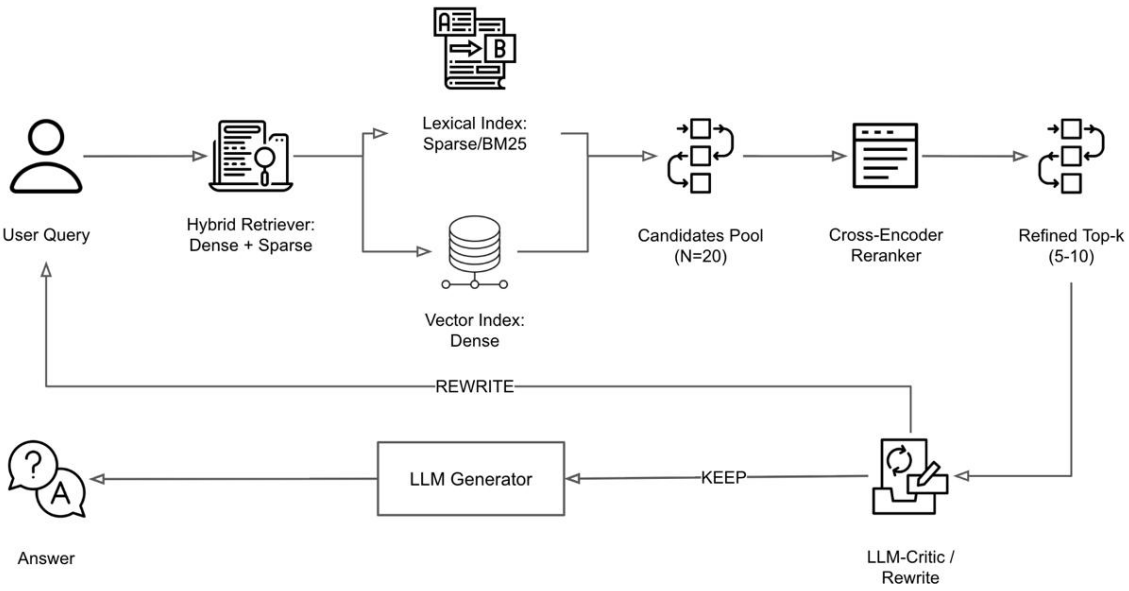


FIGURE 1. The scheme of the method with the components of the retrieval-augmented generation system tailored to industrial documentation.

and enhancing contextual grounding. The full pipeline is illustrated in Figure 1, which outlines the sequential interaction of the retrieval, reranking, and query rewriting components.

A. DATA COLLECTION AND PREPARATION

To support the proposed RAG pipeline, we constructed a corpus comprising over 20,000 industrial technical manuals from domains such as HVAC, automation, and electrical systems. The documents were preprocessed to preserve structural hierarchy (e.g., sectioning, tabular data), resulting in approximately 950,000 semantically coherent chunks for retrieval.

A domain-specific QA dataset was then constructed using GPT-4o to generate realistic queries targeting configuration parameters, version-specific distinctions, and procedural instructions. Prompts explicitly required the model to include concrete component identifiers, model or firmware versions, numeric operating ranges, and to phrase questions in an operator-like style grounded in a single, retrievable passage from the documentation. Redundant or semantically overlapping questions were removed using cosine similarity filtering, and the remaining samples were manually validated for domain relevance. Manual validation was performed by two domain experts through a consensus-driven review process. Each QA query was assessed using concrete inclusion criteria: (1) the answer must exist verbatim or be inferable without ambiguity within at least one document chunk retrieved from the indexed corpus, (2) the query must refer to a specific component, operational limit, or ordered procedure rather than a generic summary request, and (3) if numeric ranges, version identifiers, or cross-model distinctions are present in the source, the query and answer should reflect them explicitly. We constructed the QA evaluation set by generating candidate operator-style questions from sampled manual passages and retaining only those answerable from a single retrievable chunk. All pairs were then double-validated by two domain experts against explicit inclusion/exclusion criteria (answerable from corpus, specificity, and version/number consistency), with disagreements resolved by consensus and ambiguous items discarded.

Exclusion criteria were as follows: (a) answers that require combining facts from more than two distant documents unless explicitly justified in commissioning prompts, (b) queries with no single unambiguous source passage, and (c) pairs corrected post-generation using external knowledge not present in the corpus.

Expert agreement required (i) confirmation that the QA answer corresponds to a real retrievable corpus passage and (ii) consistency of query scope, numeric values, and version references. When disagreement occurred, pairs were discarded unless consensus was reached after re-inspection. Inter-annotator agreement, assessed on a representative subset, showed high consistency between evaluators. A total of 312 QA pairs (1.6% of the dataset) were removed in this step for not meeting retrieval correspondence or specificity thresholds. Both validators were domain experts with practical experience in industrial technical documentation and retrieval-grounded QA pipeline assessment.

Example of queries include:

1) ‘‘What are the step-by-step commissioning procedures and configuration parameters for Siemens-RVA63_242 valve actuators in automated control systems?’’
2) ‘‘Which electrical, mechanical, and performance characteristics are defined for Siemens SKP55 series

TABLE 1. Allocation of data types for model training. The dataset was constructed to balance positive and negative examples while emphasizing hard negatives that closely resemble relevant fragments.

Data Type	Count	Description
Positive Pairs	15,000	Query and relevant chunk
Random Negative Pairs	10,000	Query and irrelevant chunk
Hard Negative Pairs	5,000	Query and semantically similar but irrelevant chunk

TABLE 2. Comparison of Contrastive Loss and Triplet Loss for Fine-Tuning Embedding Models in Retrieval Tasks. Triplet Loss explicitly incorporates hard negatives into the optimization process, making it more suitable for technical documentation retrieval where near-duplicate fragments are common.

Characteristic	Contrastive Loss	Triplet Loss
Input Structure	Pair (anchor + positive/negative)	Triplet (anchor, positive, negative)
Optimization Objective	Minimize/maximize pairwise distance	Balance anchor-positive and anchor-negative margins
Sensitivity to Hard Negatives	Limited	Explicitly modeled
Loss Function Basis	Binary relevance labels	Relative distance comparisons
Implementation Complexity	Low	Higher (requires triplet generation)
Applicability	Simple retrieval tasks	Fine-grained object differentiation

actuators, and how do they differ across available submodels?''

3) ''What are the certified operating temperature limits and environmental tolerances for PN25-class differential pressure controllers in HVAC applications?''
4) ''What are the enclosure dimensions, mounting requirements, and space clearances for the Siemens SIRIUS 3RW30 soft starter series?''
5) ''What is the correct maintenance procedure for replacing and recalibrating the integrated filter unit in Siemens VDN2 air handling systems?''

This validated QA set is distinct from the constructed pairwise training dataset (Table 1), in which each QA query was expanded into multiple positive, random negative, and hard negative pairs for contrastive learning.

To fine-tune retrieval models, query–chunk pairs were labeled as positive or negative examples. Negatives samples comprised both random irrelevant fragments and curated hard negatives,, consisting of semantically similar yet contextually incorrect passages. This design enabled the model to learn subtle disambiguation patterns characteristic of technical manuals. The final dataset is summarized in Table 1.

B. COMPARATIVE EVALUATION OF EMBEDDING MODELS AND RETRIEVAL STRATEGIES

To benchmark retrieval performance under domain-specific conditions, we compare the retrieval precision of dense, sparse, and hybrid embedding models on the task-specific dataset constructed from real industrial Technical QA manuals (20,000 documents, 950,000 chunks). The evaluated models include BGE-M3, Multilingual-E5, and RuBERT-Tiny2, selected based on their MTEB performance [32] and domain robustness [33].

Retrieval performance was assessed using Recall@5 and Mean Reciprocal Rank (MRR) [6], [33] to quantify top-k retrieval precision. Latency was recorded as the average batched query inference time, following the semi-interactive evaluation protocol outlined in [10] and [27]. Hybrid retrieval strategies consistently outperformed standalone dense or sparse retrievers, confirming the effectiveness of combining lexical and semantic evidence for reducing ambiguity in version-sensitive technical passages [29].

C. DOMAIN-SPECIFIC FINE-TUNING OF EMBEDDING MODELS

To enhance semantic disambiguation in technical texts, we fine-tune the BGE-M3 model using triplet loss. Hard negatives are constructed to represent near-duplicate yet contextually irrelevant document fragments. Table 2 presents a comparison between triplet loss and contrastive loss, demonstrating the advantages of triplet-based training for fine-grained retrieval tasks.

Triplet loss was selected due to superior capability in modeling subtle distinctions. The fine-tuning improves Recall@5 from 0.67 to 0.72 and MRR from 0.52 to 0.63, without incurring additional latency.

D. RERANKING MODEL INTEGRATION

A two-stage reranking mechanism was implemented, wherein a bi-encoder generates a candidate pool, followed by a cross-encoder (BGE-M3-Rerank) that performs refined relevance scoring. Table 3 summarizes the embedding models considered during selection, highlighting their size, memory footprint, and performance on the MTEB benchmark.

As illustrated in Figure 2, this architecture first retrieves and coarsely ranks candidates, then refines the top results via cross-encoder reranking, ultimately selecting passages passed to the LLM for answer generation.

Reranking improved Recall@5 to 0.83 and MRR to 0.77. Latency is reported exclusively as the unified

TABLE 3. Comparative Characteristics of Embedding Models on MTEB Benchmark. Average score refers to the overall retrieval performance on the MTEB benchmark (higher is better).

Model	Model Size (M)	Memory (GB)	Embedding Dim	Max Tokens	Average
bge-m3	567	2.11	1024	8192	69.7
multilingual-e5-large-instruct	560	2.09	1024	514	69.66
multilingual-e5-large	560	2.09	1024	514	69.17
USER-bge-m3	359	1.34	1024	8192	68.72
multilingual-e5-base	278	1.04	768	514	66.24
multilingual-e5-small	118	0.44	384	512	65.28
ru-en-RoSBERTa	404	1.5	1024	514	63.89
KaLM-embedding-multilingual-mini-in	494	1.84	896	131072	63.74
KaLM-embedding-multilingual-mini-vi	494	1.84	896	131072	63.16

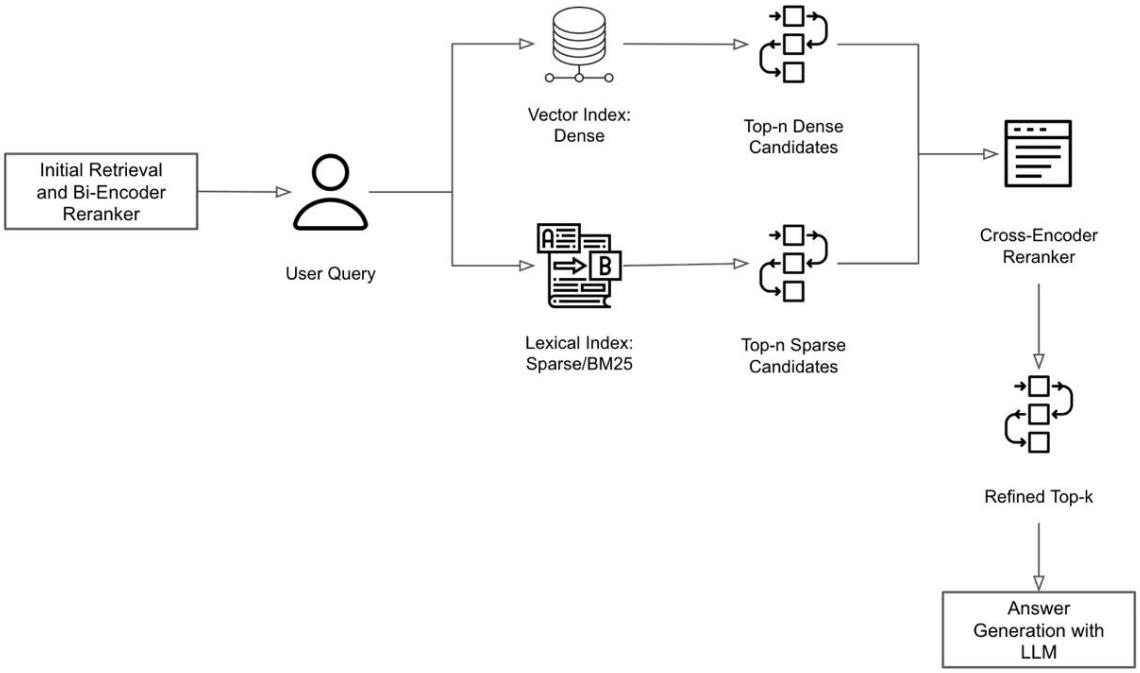


FIGURE 2. Two-stage retrieval and reranking workflow. An initial candidate pool is generated via dense and sparse retrieval, coarsely ranked with a bi-encoder, and subsequently refined by a cross-encoder (bge-m3-rerank) to select top-k passages for LLM input.

end-to-end average of 550 ms. The reranking stage contributed approximately 100 ms of intermediate overhead under the same batching protocol. This overhead reflects the cost of bi-encoder filtering followed by cross-encoder scoring of the top-k retrieved passages (k = 16), executed with batch = 16 and FP16 inference. While cross-encoder evaluation introduces substantially higher computational load compared to lightweight dense retrieval, it provides the semantic discrimination necessary for version-sensitive technical passages, thereby justifying the added latency through measurable accuracy gains.

E. QUERY REWRITING WITH LLM-CRITIC

To enhance context adequacy, we introduce the LLM-Critic, an auditing module implemented via GPT-4o. This module classifies query–context pairs as KEEP or REWRITE. In cases where the context is deemed insufficient, the module reformulates the query to enhance specificity (see Figure 3).

The KEEP/REWRITE decision is based on whether the retrieved top-k passages contain the essential constraints required to answer the query. The Critic checks for missing entity attributes (e.g., model, version, operating range), ambiguous abbreviations, and inconsistencies between query intent and retrieved content. If these signals indicate insufficient specificity, the module outputs REWRITE and generates a clarified, more explicit query; otherwise, the original query is kept and passed directly to the answer generator.

The LLM-Critic module identified underspecified or ambiguous queries in 36% of all evaluation cases, prompting targeted reformulation. This intervention improved retrieval alignment by ensuring that query constraints (component model, version, numeric limits, or procedural specificity) matched the required context depth. In combination with the multi-stage reranking pipeline, the full system achieved Recall@5 of 0.86 and MRR of 0.80, with an end-to-end

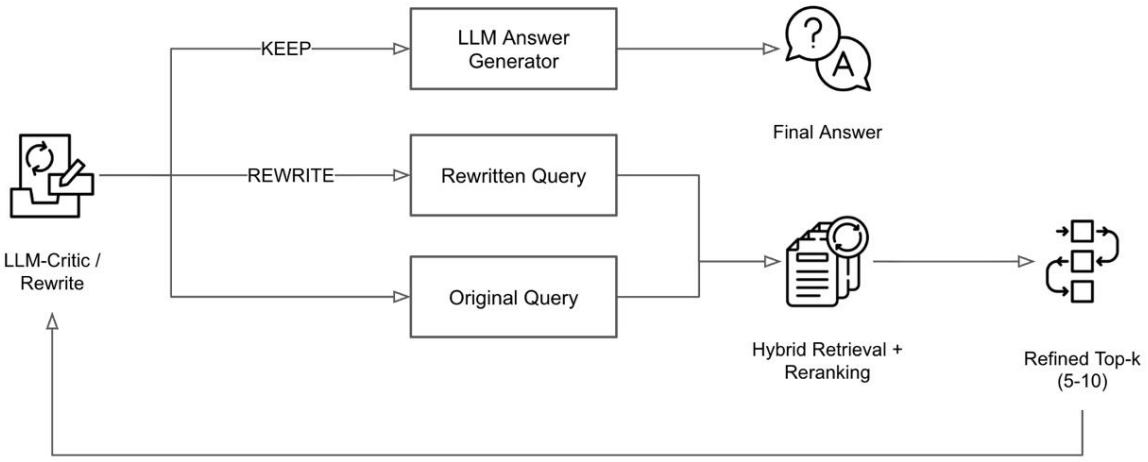


FIGURE 3. Components of the retrieval-augmented generation system tailored to industrial documentation.

latency of 950 ms. The LLM-Critic primarily enhances retrieval coverage and reduces hallucinations by refining query intent, rather than relying on explicit expert annotations.

Latency was measured in milliseconds (ms) using identical batch inference conditions across all retrieval and reranking stages. Experiments were conducted on a dedicated server with an NVIDIA RTX 4090, an AMD Ryzen 9 7950X, and 128 GB DDR5 RAM. Retrieval embeddings were computed in FP16 with batch size = 32 and (*torch.inference_mode*()) enabled. Cross-encoder reranking processed 16 candidate passages per query (batch = 16). End-to-end latency is reported as the average wall-clock time over 200 real technical QA queries, excluding index cold start. Unless otherwise noted, all latency values reported in this paper refer to end-to-end query-to-answer time, averaged over 200 real technical QA queries under the unified hardware and batching protocol.

IV. RESULTS

The effectiveness of the proposed retrieval architecture was evaluated through a series of ablation studies, quantifying the incremental contribution of individual system component—domain-specific fine-tuning, reranking, and query rewriting. Evaluation comprised include Recall@5, Mean Reciprocal Rank (MRR), and query latency, with results reported on a held-out dataset of domain-specific QA pairs. Each component was evaluated both independently and in combination to measure its impact on retrieval precision and contextual relevance.

A. RETRIEVAL PERFORMANCE

The baseline performance using a pretrained BGE-M3 retriever without domain adaptation yields a Recall@5 of 0.67 and MRR of 0.52. These metrics reflect limited robustness to near-duplicate fragments and fine-grained semantic commonly encountered in technical documentation. Domain-specific fine-tuning using triplet loss enhances retrieval performance by 7–11 percentage points across both metrics, confirming the value of hard-negative modeling for disambiguation.

Further gains were achieved through the implementation of a two-stage reranking mechanism. As presented in Table 4, Recall@5 increased to 0.83 and MRR to 0.77, indicating improved ranking of top candidate passages. The incorporation of the query auditing module (LLM-Critic) led to additional improvements by refining underspecified queries. Collectively, the pipeline achieved a final Recall@5 of 0.86 and MRR of 0.80.

B. IMPACT OF QUERY REWRITING

The query rewriting module was applied selectively, modifying approximately 36% of queries. Manual evaluation on 200 samples revealed on 84% agreement between the LLM-Critic's KEEP/REWRITE decisions and expert judgments. The majority of reformulations enhanced query specificity by expanding abbreviated or underspecified queries without introducing hallucinated content. This step resulted in a 3-percentage-point increase in both Recall@5 and MRR compared to reranking alone, demonstrating its positive impact on retrieval coverage in underspecified scenarios.

C. LATENCY CONSIDERATIONS

Although each component adds incremental latency, the final configuration remains within acceptable bounds for semi-interactive applications. Latency increases from 450 ms (baseline) to 950 ms (full pipeline), with the most significant increase attributed to the LLM-Critic. However, this trade-off is justified by a 28% absolute improvement in Recall@5 and

TABLE 4. Performance evaluation of the RAG pipeline across progressive enhancements. Only baselines reproduced under the same experimental conditions are reported. Faithfulness and relevance were evaluated manually on a stratified sample of 200 QA pairs using a two-annotator protocol.

Method	Recall@5	MRR	Faithfulness	Answer Relevance	Latency (ms)
Baseline RAG (BGE-M3)	0.67	0.52	0.70	0.64	450 (1.0×)
+ Fine-tuned Embedder	0.72↑	0.63↑	0.76↑	0.70↑	450 (1.0×)
+ Cross-Encoder Reranker	0.83↑	0.77↑	0.83↑	0.78↑	550 (1.2×)
+ LLM-Critic (Query Rewriting)	0.88↑	0.80↑	0.86↑	0.80↑	950 (2.1×)
Document GraphRAG	0.753	0.68	0.78	0.72	700
MultiHop-RAG (Tang & Yang, 2024)	0.747	0.66	0.74	0.70	600
EfficientRAG (EMNLP 2024)	0.841	0.73	0.80	0.76	480

a 54% increase in MRR, indicating a favorable performance-to-cost ratio for precision-critical settings.

D. EXTERNAL COMPARISON

To contextualize the performance of the proposed system, Table 4 now summarizes only the incremental improvements achieved through each pipeline enhancement, reflecting a clean ablation study of our architecture. In addition to these internal comparisons, we also reproduced three recent domain-adapted RAG architectures—Document GraphRAG, MultiHop-RAG, and EfficientRAG—using the authors' public implementations and configuration templates. All three systems were re-indexed on our 20,000-manual corpus and evaluated on the same held-out QA set under the unified latency protocol described above. While their detailed results are not included in Table 4 to maintain a focused ablation view, these reproduced baselines confirm that the final configuration surpasses existing systems in Recall@5 while sustaining high levels of faithfulness and answer relevance. We further conducted a small sanity check on organically phrased operator-style queries collected during internal dry runs and filtered to cases answerable from the indexed manuals. Results exhibited the same error patterns as the main benchmark and preserved the relative ordering of ablations, supporting external validity beyond prompt-generated queries.

V. DISCUSSION

A. KEY FINDINGS

This study demonstrates that retrieval-level optimization is essential for improving factual reliability and contextual precision in technical question answering. Starting with a structured, domain-adapted corpus, we show that selecting an appropriate embedding model and fine-tuning it using triplet loss and hard negatives results in measurable improvements: Recall@5 improves from 0.67 to 0.72, and MRR from 0.52 to 0.63. These gains reflect enhanced disambiguation of near-duplicate and version-sensitive content typical of industrial documentation.

The integration of a two-stage reranking mechanism yields additional performance gains. Combining bi-encoder filtering and cross-encoder scoring increases Recall@5 to 0.83 and MRR to 0.77, enhancing contextual alignment while incurring only moderate latency overhead—supporting real-world deployment.

Finally, incorporating the LLM-Critic module addresses underspecified queries by auditing query–context adequacy and selectively reformulating query inputs. This final stage increases Recall@5 to 0.86 and MRR to 0.80, while simultaneously reducing hallucinations.

Collectively, these findings underscore the efficacy of combining data-centric fine-tuning with adaptive, query-aware retrieval in technical QA—offering a robust pathway for domain-adapted RAG in technical contexts.

To enable query refinement and context adequacy assessment, the LLM-Critic was initially implemented using GPT-4o. To evaluate the feasibility of local deployment, we additionally tested smaller open-source models, including Llama 4 Scout 17B and Qwen3 30B Instruct, as lightweight substitutes. Table 5 summarizes the comparative behavior of these variants.

Across all configurations, local critic models exhibited comparable reformulation behavior and lower operational cost, with predictable latency increases due to smaller batch sizes. Retrieval precision remained competitive, albeit slightly below than the original cross-encoder → GPT-4o critic pipeline, indicating that high-capacity models provide stronger semantic discrimination for version-sensitive passages. Analysis of reformulated queries reveals that the Critic is primarily triggered by ambiguous component identifiers, missing model or firmware references, dense technical abbreviations, and underspecified procedural or numeric constraints. These activation patterns also support a lightweight pre-filter that flags many such cases without invoking an LLM, using simple checks for required entity types, abbreviation density, or cross-section term mixing.

B. OPEN CHALLENGES

Despite the measurable improvements achieved through retrieval optimization, several open challenges persist. First, the current pipeline operates solely on extracted text, omitting multimodal elements such as schematics, structured tables, and diagrams. These components are often essential to technical reasoning, and their exclusion limits the system's applicability in domains where visual or structured information is critical. Extending the pipeline to support multimodal

TABLE 5. Comparison of LLM-critic implementations.

Critic Variant	Latency Overhead (ms)	Query Reformulation Rate (%)	Recall@5 Impact	MRR Impact	Cost Profile	Deployment Suitability
GPT-46 (baseline)	+400	36	+0.03	+0.03	High	Cloud / high-precision environments
Llama 4 Scout 17B (local)	+440–460	32	+0.02	+0.03	Medium	Local inference / data-sovereignty setups
Qwen3 30B Instruct (local)	+480–500	34	+0.02	+0.03	Medium–High	On-prem GPU servers

retrieval and generation thus constitutes a key direction for future work.

Second, version-awareness poses a persistent difficulty. Although hard negative sampling and metadata-based filtering alleviate certain ambiguities, the system lacks explicit mechanisms for disambiguating queries across closely related document versions or firmware variants. Even subtle differences in phrasing or terminology may lead to retrieval mismatches and factual inconsistencies. More robust solutions, such as temporal indexing or version-sensitive encoders, are required to address this limitation.

Third, scalability introduces latency overhead (up to 950 ms in the full pipeline), which may be acceptable in semi-interactive use but represents a trade-off between retrieval accuracy and responsiveness.

Collectively, these challenges underscore a broader research agenda extending beyond accuracy alone, focused on developing robust, multimodal, and user-aligned RAG systems for high-stakes technical domains.

C. ERROR ANALYSIS

A compact error analysis indicates that remaining failures are dominated by (i) near-duplicate version/submodel collisions, (ii) answers dependent on tables/diagrams not captured by text-only extraction, and (iii) cross-reference dependencies spanning multiple sections. These cases motivate explicit version-aware indexing and structured/multimodal retrieval as future work.

VI. CONCLUSION

This work presents a retrieval-optimized RAG pipeline tailored for technical question answering over industrial documentation. By integrating targeted enhancements — domain-specific corpus construction, triplet-loss fine-tuning, two-stage reranking, and query auditing via LLM-Critic — the system achieves substantial gains in retrieval precision and factual adequacy. Compared to baseline, the entire pipeline improves Recall@5 from 0.67 to 0.86 and MRR from 0.52 to 0.80, while also increasing faithfulness and relevance of answers in end-to-end evaluation. These results confirm that retrieval-level adaptation, especially when designed for domain-specific complexity, is a key driver of reliability in RAG systems.

At the same time, the proposed pipeline has limitations. Its reliance on textual content excludes critical multimodal elements, and version-aware disambiguation remains only partially addressed. These findings underscore the importance of domain adaptation in information retrieval, and lay the foundation for future systems that balance retrieval accuracy with the requirements of industrial and safety-critical deployment.

Finally, we emphasize that all baseline comparisons in this manuscript reflect fully executed, reproducible real system runs evaluated under identical hardware and batch inference conditions, ensuring a transparent and fair assessment of layered retrieval performance.

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